

IberRail S.A.

TECHNICAL SPECIFICATION

Passenger Door and Access Systems

IB-EMU-450 Regional Electric Multiple Unit — French RFN Cross-Border Authorisation

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French National Rule:	Arrete du 19 mars 2012, Article 49 (RFN Rolling Stock)
Subsystem:	LOC&PAS; - Passenger Rolling Stock, Door Systems
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Authorisation Scope:	French national network (RFN) — cross-border extension from existing Spain/Portugal authorisation
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Classification:	RESTRICTED - COMMERCIAL IN CONFIDENCE

This document constitutes the technical specification for the passenger door and access systems of the IberRail IB-EMU-450 regional electric multiple unit, specifically for the cross-border authorisation extension to the French national network (RFN). IberRail holds existing type authorisation for Spain and Portugal (ERA ref. ERA/2024/IB-450/AUTH-007). This revision documents the two additional compliance conflicts identified during the French RFN authorisation process and IberRail's proposed resolutions, for review by EPSF and the appointed NoBo CERTIFER under Module SB of Decision 2010/713/EU.

Document Revision History

Rev.	Date	Author	Description of Change	Approved
1.0	12 Jun 2024	J. Morales	Initial issue for Spain/Portugal authorisation. Door system specified to EN 14752:2015 and Spain NNTR door requirements.	M. Delgado
2.0	04 Sep 2024	J. Morales	Updated following BODE GmbH FAT results. Revised obstacle detection parameters. Added Portugal NNTR cross-reference.	M. Delgado
3.1	14 Mar 2026	A. Ferreira	Final release for Spain/Portugal NoBo type examination. Spain/Portugal authorisation granted ERA/2024/IB-450/AUTH-007.	M. Delgado
4.0	18 Apr 2026	J. Morales	New revision for French RFN cross-border authorisation. Sections 4.2 and 4.3 added covering two French compliance conflicts: (1) EN 14752 vs NF F31-054 obstacle detection test protocol; (2) CAS single-agent operation requirements mandated by Arrete 2012 Art. 49 with no TSI equivalent. Open issues OI-003 and OI-004 raised.	M. Delgado

1. Scope and Applicability

1.1 Vehicle and Programme

The IberRail IB-EMU-450 is a 5-car, 160 km/h regional electric multiple unit originally developed for the Spanish and Portuguese national networks. IberRail has received a contract extension from SNCF under framework agreement SNCF-RC-2025-0019 for operation of IB-EMU-450 units on French regional services (TER services in the Occitanie and Nouvelle-Aquitaine regions), including cross-border services between Spain and France via the Hendaye and Cerbere border crossings.

This document revision covers only the French RFN authorisation scope. For the Spain/Portugal authorisation technical file, refer to IB-EMU-450-TS-004 Rev. 3.1. The door system hardware is identical across all network variants; differences arise solely in the regulatory compliance assessment and in software configuration parameters within the door control unit.

1.2 French RFN Authorisation Context

IberRail holds existing vehicle type authorisation for the IB-EMU-450 on the Spanish RENFE network and the Portuguese IP network (ERA OSS ref. ERA/2024/IB-450/AUTH-007, granted 2 March 2026). The French RFN cross-border extension is sought under Article 21(14) of Directive (EU) 2016/797, which provides a simplified authorisation pathway for vehicles already authorised in another Member State, subject to verification of national rules applicable to the additional area of use.

The EPSF has identified two areas where French national rules impose requirements beyond or different from the TSI, requiring explicit compliance demonstration by IberRail:

- Area 1 (Section 4.2): Obstacle detection test protocol - the French Arrete du 19 mars 2012 Article 49 references NF F31-054, which specifies a more detailed test methodology than EN 14752:2015 (referenced in the TSI).
- Area 2 (Section 4.3): Single-agent operation (CAS) - French regional services operate in CAS mode; Arrete 2012 Art. 49 mandates NF F31-054 compliance for CAS-operated trains, imposing four functional requirements that have no equivalent in the LOC&PAS; TSI.

NOTE: This specification covers only the passenger door and access subsystem. For the complete IB-EMU-450 technical file including all other subsystems assessed for French RFN compatibility, refer to master document IB-EMU-450-TS-001 Rev. 5.0 (French RFN edition).

2. Reference Documents

Ref.	Document / Standard	Title / Description
[1]	Commission Regulation (EU) 2023/1427	LOC&PAS; TSI - Technical Specifications for Interoperability, Locomotives and Passenger Rolling Stock. OJ L 174, 7.7.2023.
[2]	Directive (EU) 2016/797	Interoperability of the rail system within the European Union.

Ref.	Document / Standard	Title / Description
[3]	Arrete du 19 mars 2012	Fixant les objectifs, les methodes, les indicateurs de securite et la reglementation technique de securite et d'interopabilite applicables sur le reseau ferre national. Published in Journal Officiel de la Republique Francaise. Article 49 covers passenger door requirements for rolling stock on the RFN.
[4]	EN 14752:2015	Railway Applications - Body Side Systems (Doors). European standard referenced in LOC&PAS; TSI Art. 4.2.3.1 for obstacle detection.
[5]	NF F31-054	Materiel ferroviaire roulant - Portes d'accès voyageurs des voitures et automoteurs - Systeme de service des portes pour trains exploites en conduite agent seul. French national standard. Referenced in Arrete 2012 Art. 49 for CAS-operated trains.
[6]	IEC 61508-3:2010	Functional Safety of E/E/PE Safety-Related Systems, Part 3.
[7]	IB-EMU-450-TS-001 Rev. 5.0	IberRail IB-EMU-450 - Master Vehicle Technical Specification (French RFN edition).
[8]	IB-EMU-450-TS-004 Rev. 3.1	IberRail IB-EMU-450 - Door System Technical Specification (Spain/Portugal authorisation - superseded for French scope).
[9]	IB-EMU-450-TEST-012 Rev. 2.0	Door System Factory Acceptance Test Report (BODE GmbH, Sep 2024). Covers EN 14752 testing for Spain/Portugal authorisation.
[10]	EPSF/2025/PR/IB-450-FR-001	EPSF Preliminary Review - French RFN authorisation, door system section. Issued 22 February 2026. Identified Areas 1 and 2 above.
[11]	IB-EMU-450-CALC-007	Kinetic energy analysis at door contact - NF F31-054 Section 6.3 compliance calculation. Rev. 1.0, March 2026.

3. Definitions and Abbreviations

Term / Abbreviation	Definition
RFN	Reseau Ferre National - the French national railway network managed by SNCF Reseau, on which IberRail seeks cross-border authorisation for the IB-EMU-450.
EPSF	Etablissement Public de Securite Ferroviaire - the French National Safety Authority (NSA) under Directive (EU) 2016/797.
CERTIFER	The appointed Notified Body for this French RFN authorisation, accredited under Directive (EU) 2016/797 for LOC&PAS; subsystem assessment (accreditation ref. FR-NB-2016/0087).
CAS	Conduite Agent Seul - single-agent operation, where one agent both drives the train and controls door service from the cab without a dedicated platform conductor. Standard mode for French TER regional services.
NF F31-054	French national standard for passenger door access systems on trains operated in CAS mode. Specifies platform surveillance, closure confirmation, and obstacle detection test requirements beyond the TSI baseline. Published by AFNOR.
EN 14752	European standard for railway body side door systems, referenced in LOC&PAS; TSI Art. 4.2.3.1. Forms the basis for IberRail's existing Spain/Portugal authorisation door compliance.
TER	Transport Express Regional - French regional rail services operated by SNCF Voyageurs under agreement with regional councils. The primary service type for IB-EMU-450 French deployment.
AMEC	Autorisation de Mise en Exploitation Commerciale - French commercial service authorisation issued by EPSF following successful type examination.
Obstacle Detection Sensitivity	The minimum diameter of a cylindrical test obstacle that the door system must detect and respond to by reversing closure, expressed in mm.
DCU	Door Control Unit - the BODE GmbH DCU-3100 safety-rated electronic controller managing door operation on the IB-EMU-450. SIL 2 certified per IEC 61508.
AW0 / AW3	AW0: empty vehicle (no passengers). AW3: crush load at 8 pass/m2. Standard loading conditions for door system testing.
MITMA	Ministerio de Transportes, Movilidad y Agenda Urbana - Spanish Ministry of Transport (relevant to existing Spain authorisation).

4. Passenger Door System - Technical Requirements

4.1 System Overview and Configuration

The IB-EMU-450 door system is unchanged from the Spain/Portugal authorisation variant. No hardware modifications are required for French RFN operation. Two software configuration parameters within the BODE DCU-3100 are set differently for the French RFN variant to satisfy NF F31-054 CAS requirements (see Section 4.3). All hardware specifications remain as documented in IB-EMU-450-TS-004 Rev. 3.1.

Parameter	Value	Reference
Door type	Sliding plug door, outward plug then slide	EN 14752 Cl. 5.2
Doors per 5-car set	24 total (8 per intermediate car, 4 per driving trailer)	IB-EMU-450-TS-001
Clear opening width	1,400 mm nominal	LOC&PAS; TSI Art. 4.2.2.4
Clear opening height	2,050 mm nominal at AW0	LOC&PAS; TSI Art. 4.2.2.4
Platform heights served	550 mm and 920 mm above top of rail (RFN platform registry)	RFN infrastructure register
Door drive	Electromechanical, 24 V DC brushless motor, integrated encoder	BODE GmbH DWG 4521-B
Door control unit	BODE DCU-3100, SIL 2 certified per IEC 61508	IB-EMU-450-TES T-012
Max. closing speed	0.35 m/s at leading edge	EN 14752 Cl. 6.4.2
Obstacle detection	Dual-zone: IR light curtain (primary) + resistive edge sensor (secondary)	Section 4.2
Operating mode (France)	CAS - Conduite Agent Seul. One agent drives and controls door service.	French RFN / NF F31-054
Emergency release	Manual inside and outside, operable without tools	LOC&PAS; TSI Art. 4.2.5.5

4.2 Door Obstacle Detection System

This section documents the first French compliance conflict identified by EPSF during preliminary review (ref. EPSF/2025/PR/IB-450-FR-001). The IB-EMU-450 door obstacle detection system was designed and tested against EN 14752:2015 for Spain/Portugal authorisation. The French Arrêté du 19 mars 2012 Article 49 references NF F31-054, which imposes a more detailed test protocol for the same 30 mm sensitivity threshold.

4.2.1 Detection System Design (unchanged from Rev. 3.1)

Layer 1 - Infrared light curtain (primary, contactless):

Dual-beam IR curtain mounted on both door leaves, 38 kHz modulated, 10 mm beam pitch, covering 50 mm to 2,000 mm above door sill. Provides continuous contactless detection throughout door travel. Failure detected within one door cycle by DCU-3100 self-test.

Layer 2 - Resistive edge sensor (secondary, contact-based):

Pressure-sensitive silicone extrusion along the full leading edge of both door leaves (2,050 mm height). Contact force above 30 N causes conductive tracks to close, triggering reversal via the DCU-3100 safety loop. The edge sensor governs the minimum detectable obstacle diameter for certification purposes.

NOTE: Both detection layers are independently monitored by the DCU-3100. Layer 1 failure triggers reduced-speed mode (0.20 m/s) until maintenance. Layer 2 open circuit triggers door locked state and crew alarm. Safe-state behaviour is unchanged between network variants.

4.2.2 Obstacle Detection Sensitivity - Design Parameter

Parameter	IberRail Design Value	Source / Rationale
Min. detectable obstacle diameter	30 mm nominal; 27 mm worst case (sensor tolerance -10%)	LOC&PAS; TSI Art. 4.2.3.1 minimum
Max. contact force at 30 mm obstacle	130 N mean across 10 FAT cycles at AW0	IB-EMU-450-TEST-0 12 Sec. 6.3
Detection response time	Max 80 ms from contact to reversal command	BODE DCU-3100 specification
Kinetic energy at obstacle contact	0.9 J by analysis (see IB-EMU-450-CALC-007)	NF F31-054 Sec. 6.3 limit: <=1.5 J
Operating temperature	-40 degC to +70 degC ambient	IEC 62498-1 European railway environment
Ingress protection	IP 65 per IEC 60529	BODE GmbH DWG 4521-B

4.2.3 Conflict 1: LOC&PAS; TSI (EN 14752) vs French Arrete 2012 (NF F31-054)

4.2.3.1 Applicable Requirements

Two regulatory instruments specify obstacle detection requirements applicable to the IB-EMU-450 for French RFN operation. Both specify 30 mm as the minimum detectable obstacle diameter, but they reference different standards with materially different test methodologies:

Instrument	Article	Standard Referenced	Test Methodology
LOC&PAS; TSI (EU 2023/1427)	Art. 4.2.3.1 'Obstacle Detection and Re-opening'	EN 14752:2015 Clause 7.2	Test at single most unfavourable position in door height. One force measurement per test. Reversal within 100 ms. 30 mm cylinder diameter minimum.

Instrument	Article	Standard Referenced	Test Methodology
French Arrete 19 mars 2012	Article 49 'Portes d'accès voyageurs sur le RFN'	NF F31-054 Section 6.3	Test at 5 defined height positions: 250 mm, 500 mm, 900 mm, 1,300 mm, and 1,600 mm above door sill. Force measured at each position with 100 N maximum. Kinetic energy limit of 1.5 J at contact. 30 mm cylinder diameter minimum.

4.2.3.2 Conflict Analysis

The 30 mm minimum obstacle detection diameter is consistent between both standards. The conflict lies entirely in test protocol stringency. NF F31-054 Section 6.3 imposes three additional requirements not present in EN 14752 Clause 7.2:

- Testing at 5 defined height positions versus EN 14752's single worst-case position. Positions 250 mm and 1,600 mm above sill are particularly demanding - at 250 mm the door leading edge approaches maximum closure speed; at 1,600 mm the sensor geometry changes due to door leaf overlap profile.
- Maximum kinetic energy at obstacle contact of 1.5 J, which constrains the closing speed profile more tightly than the TSI. This parameter has no equivalent in EN 14752. IberRail's analysis (IB-EMU-450-CALC-007) demonstrates 0.9 J at worst-case position based on measured DCU-3100 speed profile, but physical verification at all 5 positions is required by EPSF.
- Force limit of 100 N at each of the 5 test positions, versus EN 14752's single maximum force measurement of 150 N. The IB-EMU-450 achieves 130 N at the single EN 14752 test position; compliance at all 5 NF F31-054 positions requires supplementary FAT testing.

Existing FAT documentation (IB-EMU-450-TEST-012 Rev. 2.0) satisfies EN 14752 requirements and is sufficient for TSI compliance demonstration. It does not satisfy NF F31-054 Section 6.3 for French RFN authorisation because it covers only one height position and does not report kinetic energy measurements.

NOTE: NF F31-054 is a French national standard published by AFNOR. It is not freely available online. IberRail has obtained a licensed copy (AFNOR ref. NF F31-054:2008). The standard is available for purchase at www.afnor.org. CERTIFER holds a copy for assessment purposes.

4.2.3.3 IberRail Compliance Position and Resolution Plan

IberRail has evaluated two resolution paths:

- **Option A - Supplementary NF F31-054 testing (SELECTED):** Conduct supplementary FAT at BODE GmbH covering all 5 NF F31-054 height positions and reporting kinetic energy at each. Kinetic energy compliance is expected by analysis (0.9 J vs 1.5 J limit) but physical verification is required. No hardware modification expected. Test planned May 2026 (see OI-003).
- **Option B - Acceptable means of compliance (rejected as primary path):** Submit analysis to EPSF demonstrating that EN 14752 single-position testing provides equivalent safety assurance to NF F31-054 multi-position protocol. Rejected following EPSF preliminary review indication that physical NF F31-054 testing is expected for all vehicles seeking French RFN authorisation.

WARNING: As of Rev. 4.0, supplementary NF F31-054 Section 6.3 testing (V-010) has not been completed. Existing IB-EMU-450-TEST-012 satisfies the TSI and the Spain/Portugal authorisation but does NOT satisfy Article 49 of the Arrete du 19 mars 2012 for French RFN operation. Type examination for French RFN cannot be finalised until V-010 results are submitted to CERTIFER.

4.3 Single-Agent Operation Door System (CAS - Conduite Agent Seul)

This section documents the second French compliance conflict identified by EPSF. The LOC&PAS; TSI contains no specific requirements for single-agent-operated door systems. The French Arrete du 19 mars 2012 Article 49 mandates conformity with NF F31-054 for all trains operated in CAS mode on the RFN. This creates four functional requirement gaps that must be resolved for French authorisation.

French TER regional services, for which the IB-EMU-450 is contracted under SNCF-RC-2025-0019, operate exclusively in CAS mode. One agent drives the train and manages all door service operations from the driving cab without a dedicated conductor on the platform. This is the standard operational model for French regional services and is not a special case.

4.3.1 CAS Requirements from NF F31-054 - Gap Analysis

NF F31-054 establishes a comprehensive functional specification for CAS door systems. The table below identifies each NF F31-054 CAS requirement, its TSI equivalent (if any), and the gap that IberRail must address:

NF F31-054 CAS Requirement	Sec .	TSI Equivalent	Gap Assessment	IberRail Response
Platform surveillance: the agent must have visual confirmation that the platform is clear before issuing door closure command.	4.2	None. TSI Art. 4.2.3 covers door closure behaviour but contains no requirement for platform visibility from cab.	FULL GAP - no TSI equivalent. Requirement entirely absent from LOC&PAS; TSI.	CCTV platform surveillance system installed. See Section 4.3.2.
Closure confirmation interlock: door closure command is not accepted by the door system until the agent has actively confirmed platform clearance via a dedicated two-step control sequence in the cab.	4.3	None.	FULL GAP. TSI permits single-step door close command.	DCU-3100 CAS_CONFIRM parameter enabled for French variant. Two-step sequence required: observe CDU, then press CAS confirm, then issue door close command.
Passenger alarm in CAS mode: activation must trigger audible and visual alert in cab AND lock all doors open until acknowledged by agent.	5.1	TSI Art. 4.2.5.3 requires passenger alarm to alert driver. Does not specify door lock-open behaviour or acknowledgement requirement.	PARTIAL GAP. TSI is less prescriptive. French requirement adds door lock-open and active agent acknowledgement.	DCU-3100 CAS_ALARM_LOCK parameter set TRUE for French variant. Doors remain locked open until agent acknowledges via cab control.

NF F31-054 CAS Requirement	Sec .	TSI Equivalent	Gap Assessment	IberRail Response
Re-closure dwell time: after obstacle detection reversal in CAS mode, minimum 5-second pause before automatic re-closure attempt is permitted. Agent must re-confirm platform clearance before re-closure.	5.2	TSI Art. 4.2.3 specifies reversal on detection but imposes no minimum dwell time before re-closure.	PARTIAL GAP. TSI permits immediate re-closure attempt after reversal.	DCU-3100 CAS_RECLOSE_DWELL parameter set to 5 s for French variant (Spain/Portugal variant uses 3 s default).

4.3.2 Platform Surveillance System

To satisfy NF F31-054 Section 4.2, IberRail has specified a cab-mounted digital CCTV platform surveillance system as an RFN-variant fitment on all IB-EMU-450 units deployed on French TER services. This system is not fitted on Spain/Portugal-variant units. The system comprises:

- 4 fixed IP cameras per car side (8 per car, 40 per 5-car set), mounted at door lintel height (2,100 mm above sill), providing continuous coverage of the full platform zone adjacent to each door. Camera field of view covers 3 m platform depth at 550 mm and 920 mm platform heights.
- Cab display unit (CDU) integration: a dedicated 10-inch touchscreen display in the driving cab shows a 4-panel composite view of the platform-side cameras on both sides of the train simultaneously. The display activates automatically when the train decelerates below 30 km/h approaching a station stop.
- CAS confirm button: a dedicated illuminated pushbutton on the cab desk (separate from the door close command button) that the agent must press after visually confirming platform clearance on the CDU. The DCU-3100 will not accept a door close command unless the CAS confirm button has been pressed within the preceding 30 seconds.
- Automatic inhibit: if the CAS confirm sequence is not completed within 45 seconds of the door open command, an audible warning activates in the cab and is repeated every 15 seconds. The warning does not prevent door closure but is logged to the TCMS with a timestamp.

NOTE: The CCTV platform surveillance system is an RFN-specific variant feature supplied by Bosch Security Systems under sub-contract IB-SUB-2025-BOSCH-002. The system has been integrated with the existing BODE DCU-3100 via the MVB interface. No modification to the door hardware or the DCU-3100 hardware is required; all CAS behaviour is implemented via DCU-3100 software parameter configuration.

4.3.3 DCU-3100 Software Configuration for French RFN Variant

The following DCU-3100 software parameters differ between the Spain/Portugal authorisation variant and the French RFN variant. All parameters are stored in the DCU-3100 non-volatile configuration register and set during vehicle commissioning. Parameter integrity is verified at each power-on via DCU self-test.

DCU-3100 Parameter	Spain/Portugal Value	French RFN Value	NF F31-054 Basis
CAS_MODE	FALSE	TRUE	Sec. 4.1 - activates all CAS functional requirements
CAS_CONFIRM_REQ	FALSE	TRUE	Sec. 4.3 - enables two-step closure confirmation

DCU-3100 Parameter	Spain/Portugal Value	French RFN Value	NF F31-054 Basis
CAS_ALARM_LOCK	FALSE	TRUE	Sec. 5.1 - door lock-open on passenger alarm in CAS mode
CAS_RECLOSE_DWELL	3 s	5 s	Sec. 5.2 - minimum dwell before re-closure after detection
CAS_CONFIRM_TIMEOUT	N/A	30 s	Sec. 4.3 - maximum interval between CAS confirm and door close
CDU_AUTOACTIVATE_SPD	N/A	30 km/h	Sec. 4.2 - speed below which CDU platform view auto-activates

4.3.4 NF F31-054 Conformity Assessment

NF F31-054 CAS requirements fall outside the scope of the LOC&PAS; TSI Module SB conformity assessment. CERTIFER must assess NF F31-054 conformity as a separate activity under a bilateral agreement with IberRail (ref. CERTIFER/2025/IB-450/NF-F31-054). This assessment covers the four CAS functional requirements identified in Section 4.3.1 and the platform surveillance system specification in Section 4.3.2.

CERTIFER's NF F31-054 preliminary assessment (interim report issued 15 March 2026) confirmed that the hardware and software design is broadly compliant but identified two open items requiring resolution before the final assessment report can be issued (see OI-004).

WARNING: NF F31-054 final conformity assessment by CERTIFER is pending resolution of OI-004 items. EPSF cannot issue the AMEC for French RFN operation until CERTIFER's final NF F31-054 assessment report is received and accepted. IberRail targets AMEC issuance by September 2026 to meet SNCF commercial service entry deadline.

4.4 Door Gap, Cycle Time, and Force Limits

The following parameters are unchanged from the Spain/Portugal authorisation and apply equally to the French RFN variant. No conflict exists between EN 14752 and NF F31-054 on these parameters.

Parameter	Requirement	IberRail Design Value	Reference
Max. door gap (fully closed)	≤ 10 mm	6.5 mm nominal	LOC&PAS; TSI Art. 4.2.3.2
Gap variation across height	≤ 5 mm	3.0 mm	EN 14752 Cl. 6.3
Max. door opening time	≤ 8 s	4.2 s mean FAT	EN 14752 Cl. 6.4.1
Max. door closing time	≤ 12 s	5.8 s mean FAT	EN 14752 Cl. 6.4.1
Max. closing force (30 mm)	≤ 150 N	130 N max FAT	EN 14752 Cl. 7.3 / NF F31-054 Sec. 6.3
Max. kinetic energy at contact	≤ 1.5 J	0.9 J analysis	NF F31-054 Sec. 6.3 (physical V-010 pending)
Door interlock release speed	≤ 5 km/h	3 km/h (DCU)	Traction interlock

4.5 Emergency Release

Each door leaf is fitted with internal and external emergency release handles, operable without tools per LOC&PAS; TSI Article 4.2.5.5. In CAS mode, emergency release activation is detected by the DCU-3100 and reported to the cab display with an audible alert and a door-specific fault code. This behaviour satisfies NF F31-054 Section 5.3 requirements for CAS emergency release monitoring and is unchanged from the Spain/Portugal authorisation variant.

5. Interface Requirements

Interface	Signal / Data	Protocol	Document
Train Control & Monitoring (TCMS)	Door status, detection events, CAS confirm status, CAS alarm events, cycle counters, configuration variant flag	MVB per IEC 61375-1	IB-EMU-450-TS-010
Cab Display Unit (CDU)	Platform camera composite view (French variant), CAS confirm button, door fault alerts, CAS alarm display	Ethernet + MVB via TCMS	IB-EMU-450-TS-011
Traction & Braking System	Door interlock - all doors closed and locked (traction permissive). CAS confirm status (advisory, not safety-critical interlock).	Hardwired 24 V DC safety loop + MVB redundant	IB-EMU-450-TS-015
Passenger Alarm System	Alarm activation signal, CAS door lock-open command (French variant), acknowledgement signal from cab	DCU-3100 auxiliary I/O, 24 V DC + MVB	IB-EMU-450-TS-016
Platform CCTV System (French variant only)	Camera feed to CDU, camera health status to TCMS	Ethernet GbE, IP cameras	IB-EMU-450-TS-017-FR
Vehicle Diagnostic Recorder	All fault codes, CAS events, detection events, configuration variant identifier, 90-day rolling buffer	Ethernet, IEC 61375-3-4 (ETB)	IB-EMU-450-TS-020

6. Verification and Validation Plan

ID	Activity	Method	Applies To	Status
V-001	Obstacle detection - 30 mm cylinder, EN 14752 single position	FAT at BODE GmbH per EN 14752 Cl. 7.2	Sec. 4.2.2, TSI Art. 4.2.3.1	COMPLET E - TEST-012 Sec. 6.3
V-002	Door gap - 10 mm limit	Gauge measurement, 5 points per door leaf	Sec. 4.4	COMPLET E - max 6.8 mm
V-003	Door cycle time - open and close	Timing measurement, 20 cycles at AW0	Sec. 4.4	COMPLET E - TEST-012
V-004	SIL 2 assessment - DCU-3100	FMEA + FMECA per IEC 61508-3	Section 4.1	COMPLET E - cert. SIL -DCU-2024-008
V-005	Emergency release - operability	Human factors evaluation, 5 operators	Sec. 4.5	COMPLET E - TEST-012
V-006	CAS DCU-3100 parameter configuration verification	Software configuration audit by CERTIFER assessor	Sec. 4.3.3	COMPLET E - CERTIFER interim report
V-007	Platform CCTV system - cab visibility assessment at RFN platforms	On-track assessment at 550 mm and 920 mm platform heights	Sec. 4.3.2	COMPLET E - CERTIFER interim report
V-008	CAS confirm interlock - functional test	Functional test per NF F31-054 Sec. 4.3 procedure	Sec. 4.3.1	COMPLET E - CERTIFER interim report
V-009	Passenger alarm CAS lock-open behaviour	Functional test, 10 alarm activations in CAS mode	Sec. 4.3.1	COMPLET E - CERTIFER interim report
V-010	Obstacle detection - 30 mm cylinder NF F31-054 protocol (5 height positions, kinetic energy measurement)	Supplementary FAT at BODE GmbH per NF F31-054 Sec. 6.3	Sec. 4.2.3, OI-003	PENDING - planned May 2026

ID	Activity	Method	Applies To	Status
V-011	NF F31-054 full conformity assessment - CERTIFER final report	Document review + on-vehicle assessment by CERTIFER	Sec. 4.3.4, OI-004	IN PROGRESS - report expected Jul 2026

7. Open Issues and Derogation Requests

ID	Description	Impact	Resolution Plan	Target
OI-001	Spain authorisation: door obstacle detection at 30 mm design vs Spain NNTR 50 mm requirement. BODE GmbH ECR-2026-0041 raised for revised resistive edge sensor extrusion. (Carried from Rev. 3.1 - Spain authorisation scope.)	CRITICAL for Spain - does not affect French RFN scope	BODE GmbH sensor retrofit planned April 2026. Separate from French authorisation track.	30 Apr 2026
OI-002	Spain authorisation: derogation application under Art. 7(4) to MITMA for interim 30 mm operation pending OI-001 retrofit. (Carried from Rev. 3.1 - Spain authorisation scope.)	MEDIUM for Spain - does not affect French RFN scope	IberRail legal team decision pending.	18 Apr 2026
OI-003	French RFN authorisation: NF F31-054 Section 6.3 supplementary obstacle detection testing at 4 additional height positions (250 mm, 500 mm, 1,300 mm, 1,600 mm) not yet conducted. Existing FAT covers EN 14752 single-position only and does not satisfy Arrete 2012 Art. 49 for French authorisation.	HIGH - blocks Conflict 1 resolution and V-010 completion	IberRail to attend BODE GmbH Hanover facility for supplementary FAT. Test to cover all 5 NF F31-054 positions plus kinetic energy measurements. Results submitted to CERTIFER within 5 days.	31 May 2026
OI-004	French RFN authorisation: CERTIFER NF F31-054 interim assessment (15 March 2026) identified two minor non-conformities in CAS mode: (a) CAS_CONFIRM_TIMEOUT parameter value of 30 s exceeds NF F31-054 Section 4.3 maximum of 25 s; (b) CAS alarm acknowledgement tone duration 2 s vs NF F31-054 Section 5.1 minimum of 3 s. Both require DCU-3100 software parameter changes.	HIGH - blocks CERTIFER final report and AMEC issuance	BODE GmbH to deliver DCU-3100 software update addressing both items by 10 May 2026. IberRail to re-test and submit to CERTIFER. Final NF F31-054 assessment report expected July 2026. AMEC target: September 2026.	Jul 2026

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